

HEAD MODEL

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1. HEAD ANATOMY

- skull
- white matter
- gray matter
- CSF

2. ENVIRONMENT

- dry air

3. MATERIAL MODELS AND PROPERTIES

3.1. **Skull.** From Taylor and Ford,¹ “The skull is represented by a compressible linear elastic perfectly plastic constitutive model and an accumulated strain-to-failure model to capture fracture, fit to data reported by Carter² for cortical bone.” Let damage variable $D : \Omega \times [0, T] \mapsto [0, 1] \subset \mathbb{R}$ be defined as

$$D(\mathbf{x}, t) \triangleq \frac{1}{\epsilon_p^*} \int_0^t d\epsilon_p(\mathbf{x}, t) \quad (1)$$

Question: Why in Paul’s paper is ϵ_p^* inside the time integral?

where

- $\epsilon_p(\mathbf{x}, t)$ is the equivalent plastic strain, continuously updated at each time step
- ϵ_p^* = critical value of equivalent plastic strain at fracture = 0.8%³
- $D = 0$ is undamaged material
- $D = 1$ is failed material

¹Taylor PA, Ford CC. Simulation of blast-induced early-time intracranial wave physics leading to traumatic brain injury. *Journal of biomechanical engineering*. 2009 Jun 1;131(6):061007.

²Carter DR. Biomechanics of bone. *Biomechanics of Trauma*. 1985 Oct 11:135-65.

³Carter *op. cit.* Note from Taylor 2009: “The equivalent plastic failure strain was determined from Carter’s data which showed a total failure strain of 0.016 (elastic plus plastic components) and a yield strain of 0.008. The plastic strain at failure is determined to be the difference between the two.”

Question: What is the strain tensor here, the linearized small strain tensor, and is additive composition of elastic and plastic assumed?

$$\boldsymbol{\epsilon} = \frac{1}{2} (\boldsymbol{\nabla} \mathbf{u} + \boldsymbol{\nabla} \mathbf{u}^T) = \boldsymbol{\epsilon}_e + \boldsymbol{\epsilon}_p \quad (2)$$

The damage degrades both the plastic yield strength

$$Y(\mathbf{x}, t) \triangleq Y_0[1 - D(\mathbf{x}, t)] \quad (3)$$

and the fracture stress

$$\sigma(\mathbf{x}, t) = \sigma_0^*[1 - D(\mathbf{x}, t)] \quad (4)$$

where

- Y_0 = yield stress for undamaged material = 95 MPa⁴
- σ_0^* = fracture stress for undamaged material = 77.5 MPa⁵

Other material properties used for the skull are

- ρ_0 = density (reference) = 1.412 g/cc
- κ = bulk modulus = 4.82 GPa
- ν = Poisson ratio = 0.22

3.2. White Matter. Per Taylor and Ford,⁶ the white and gray matter are modeled as a compressible (Really?, Isn't this better described as "nearly incompressible"?) viscoelastic material identical to Zhang.⁷ Material properties used for the white matter are

- ρ_0 = density (reference) = 1.04 g/cc
- κ = bulk modulus = 2.37 GPa (Taylor 2009)
- ν = Poisson ratio = 0.49 (Taylor 2009)
- G_0 = short-term shear modulus = 41.0 kPa (Taylor 2009)
- G_∞ = long-term shear modulus = 7.8 kPa (Taylor 2009)
- β = decay constant = 700 s⁻¹ (Taylor 2009)

3.3. Gray Matter. Material properties used for the gray matter are

- ρ_0 = density (reference) = 1.04 g/cc
- κ = bulk modulus = 2.37 GPa
- ν = Poisson ratio = 0.49
- G_0 = short-term shear modulus = 34.0 kPa (Taylor 2009)

⁴Carter *op. cit.*

⁵Carter *op. cit.*

⁶Taylor 2009 *op. cit.*

⁷Zhang L, Yang KH, King AI. Comparison of brain responses between frontal and lateral impacts by finite element modeling. *Journal of neurotrauma*. 2001 Jan 1;18(1):21-30.

- G_∞ = long-term shear modulus = 6.4 kPa (Taylor 2009)
- β = decay constant = 700 s⁻¹ (Taylor 2009)

According to Zhang⁸ “It is generally believed that brain tissue is a highly damped material; thus, a linear viscoelastic material model combined with a large-deformation theory was chosen to model brain tissue. The behavior of this material was characterized as viscoelastic in shear with the deviatoric stress rate dependent on the shear relaxation modulus, while the compressive behavior of the brain was considered elastic.”

The deviatoric response is a three-term Maxwell viscoelastic model with a time-dependent shear modulus given by

$$G(t) = G_\infty + (G_0 - G_\infty) \exp(-\beta t) \quad (5)$$

3.4. Cerebral Spinal Fluid (CSF). The CSF is modeled as a Newtonian fluid with density ρ and viscosity η similar to those of water.⁹

Material properties used for the CSF are

- ρ_0 = density (reference) = 1.00 g/cc = 1000 kg/m³
- κ = bulk modulus = 1.96 GPa
- ν = Poisson ratio = 0.50

3.5. Dry Air. The Hertel and Kerley EOS model,¹⁰ a nonlinear tabular equation-of-state (EOS) model created specifically for shock wave applications, is used to describe the dry air, composed of a mixture of N₂ (78.09%), O₂ (21.95%), and Ar (0.96%). Material properties used for the dry air are

- ρ_0 = density (reference) = 1.218e-3 g/cc
- κ = bulk modulus = 1.00998e-4 GPa
- ν = Poisson ratio = 0.50

While the tabular nature of the EOS has no stress response function similar to that of the continuum models used above, the EOS can be visualized as a general hyperspace surface with dimensions of pressure, temperature, density, and volume. For dry air at the reference density and bulk modulus listed above, and at ambient pressure and temperature conditions, the relationship between volumetric compression and pressure can be visualized in Fig. 1.

⁸Zhang *op. cit.* page 23

⁹Ommaya AK. Mechanical properties of tissues of the nervous system. *Journal of Biomechanics*. 1968 Jul 1;1(2):127IN23137-36138.

¹⁰Hertel ES, Kerley GI. CTH reference manual: The equation of state package. Sandia National Laboratories, Albuquerque, NM, Report No. SAND98-0947. 1998 Apr.

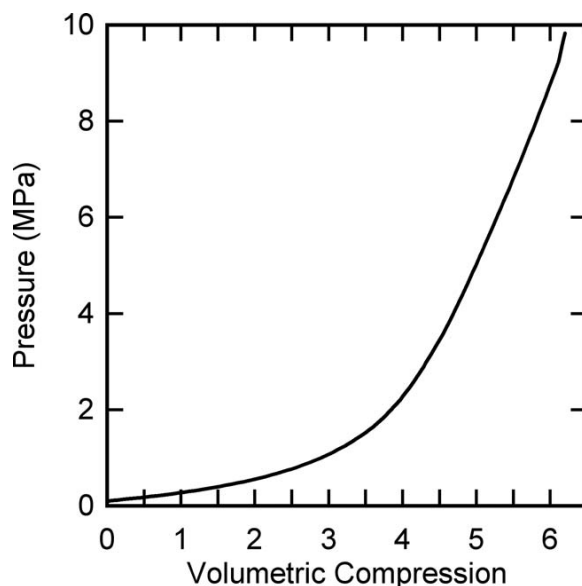


FIGURE 1. Illustration of air pressure as a function of volumetric compression at ambient conditions. (Taken from Fig. 2 of Taylor 2009.)

4. BOUNDARY CONDITIONS

4.0.1. *Energized Air.* A volume of energized air (elevated from ambient in both [internal] energy and pressure) was positioned 16 cm from the head. This produces a blast wave of 1,300 kPa (13 bars = 188.5 psi) propagating through ambient air. The pressure wave is shown in Fig. 2.

According Taylor,¹¹

Originally, blast conditions were selected consisting of 1.35 MPa (13.5 bar) peak amplitude and a pulse width of 0.6 milliseconds, which were within the marginal limits for threshold lung damage, as defined by the corrected Bowen survivability curve for blast injury [23] (see Figure 3). These conditions were chosen for two reasons. First, they were similar to those predicted to occur at a location roughly 3 metres distant from a detonated explosive device constructed from a 3 kg charge of Octol explosive. Second, they represented a limiting case for blast exposure predicted to be survivable by the Bowen lung damage criterion. However, for this blast condition, the results predicted that this strong of a blast generated intracranial stress and energy levels that were too great to be associated with mild traumatic brain injury (mTBI), the severity of brain injury displayed by this clinical TBI subject group.

Taylor continued,¹²

As an alternative, blast conditions were selected that would be less damaging to facial bone structure and more in line with conditions leading to mTBI but still close to the limits for threshold lung damage.

¹¹Taylor PA, Ludwigsen JS, Ford CC. Investigation of blast-induced traumatic brain injury. Brain injury. 2014 Jun 1;28(7):879-95.

¹²Taylor 2014 *op. cit.*

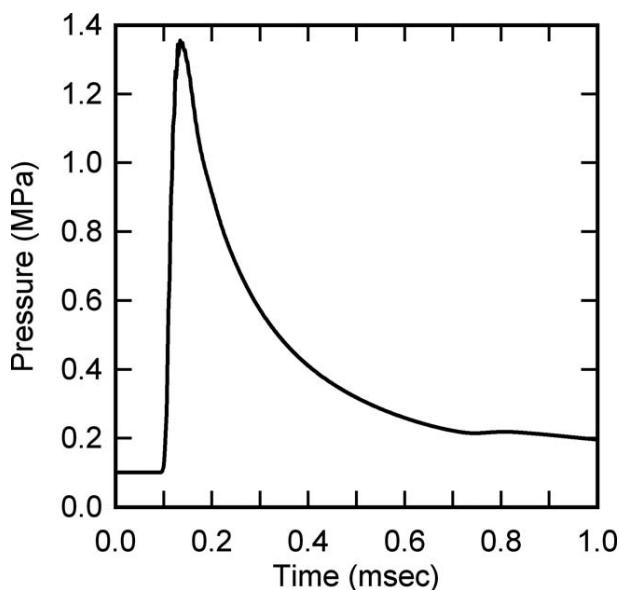


FIGURE 2. Illustration of blast air pressure pulse as a function of time (for a discrete? position in space, e.g., in front of the head?). (Taken from Fig. 3 of Taylor 2009.)

Furthermore, since this study was interested in conditions that a warfighter might experience during exposure to IED detonation, a blast history was selected that would result from a 2.3 kg charge of Composition-4 (C-4) located 2.3 metres from the head-neck model. This explosion produces an air blast of magnitude 360 KPa (3.6 bars) with a pulse width of 2.0 milliseconds as it encounters the head-neck model. A profile of this blast is displayed in Figure 4 [Fig. 3 herein]. Both the 1.35 MPa and 360 KPa blast pulses are plotted in Figure 3 [Fig. 4 herein], showing their proximity to the Bowen curve for threshold lung damage and its correction [23].

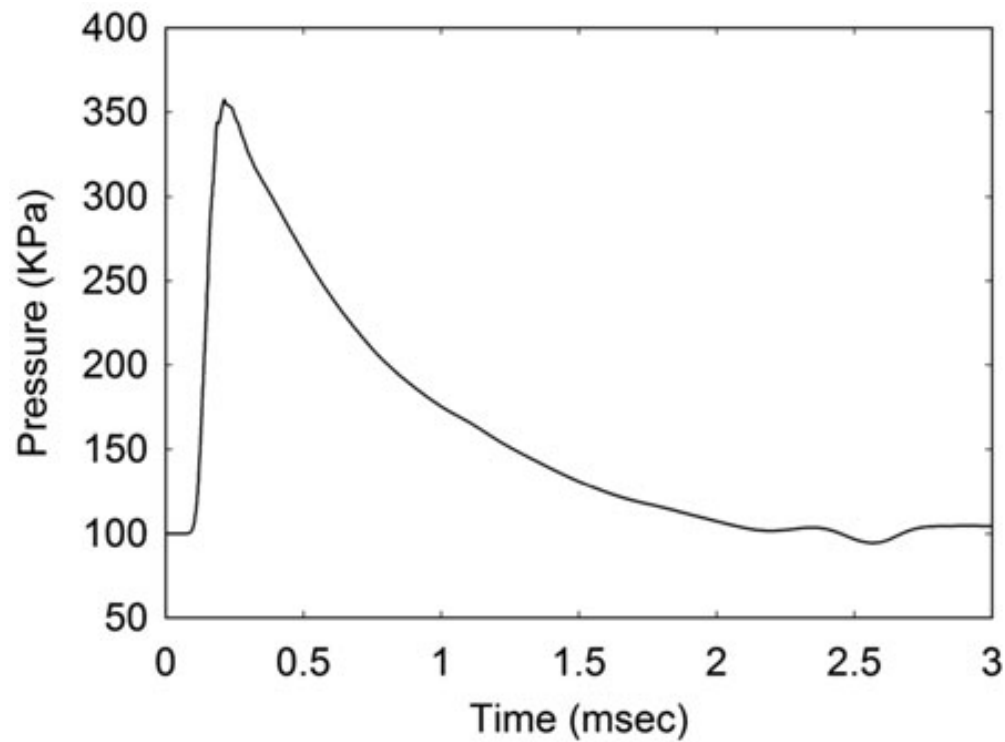


FIGURE 3. Illustration of blast air pressure pulse as a function of time (for a discrete? position in space, e.g., in front of the head?). (Taken from Fig. 4 Taylor 2014.)

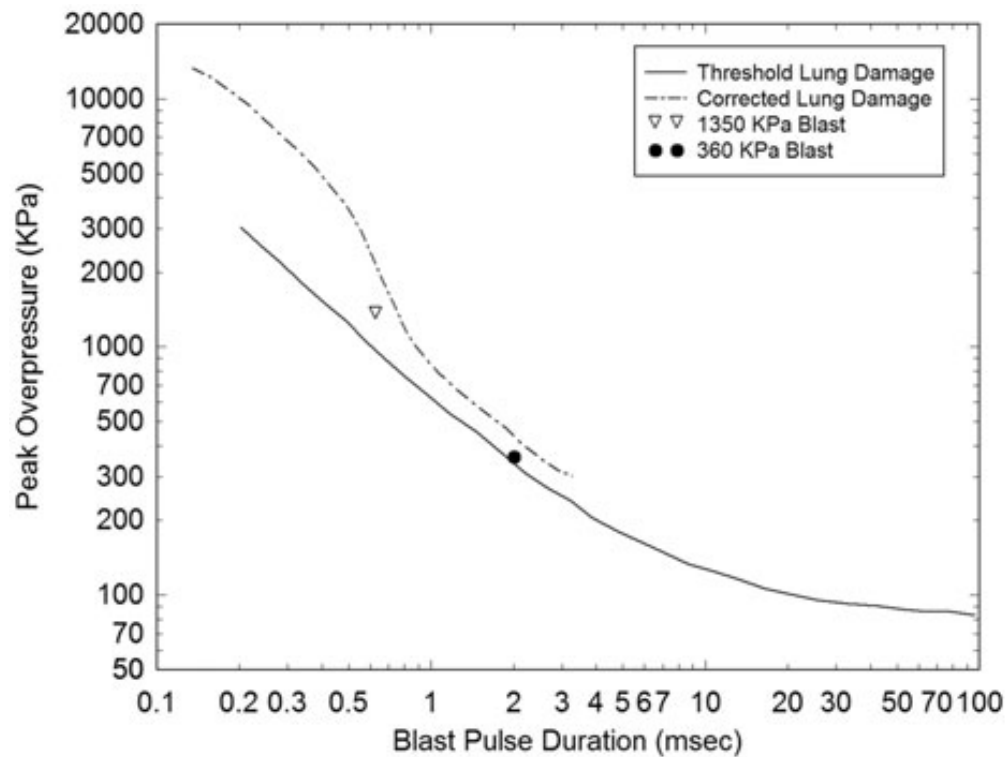


FIGURE 4. Threshold lung damage peak overpressure versus time pulse duration, with (overpressure, duration) data points shown at (1.35 MPa, 0.6 ms) and (360 kPa, 2.0 ms). (Taken from Fig. 3 Taylor 2014.)

5. CTH IMPLEMENTATION

5.1. Skull.

```

package 'Skull'
  iter 6
  material 4
  pressure 1.e6
  insert stl
    file = 'Brain_Axial_cut_skull-1.STL'
  endinsert
endpackage

*Equation-of-State (EOS) Specification (Volumetric Response)

eos
*eosfile='home/pataylo/cth_6_6_2014/SkyBSrDam2/cth/data/EOS_data'

* -----
* Mie-Gruneisen Bone
* -----
  mat4 mgr user      g0=1.0      cs=1.9838e5      cv=1.0e10
      r0=1.210      s1=1.0

From the CTH user manual, page 52:
  • cs = sound speed
  • cv = specific heat
  • g0 = Grüneisen parameter
  • r0 = initial density of Hugoniot13
  • s1 = linear coefficient of the US-UP Hugoniot curve

* -----
* Deviatoric Strength Specification
* -----

*fvp='/home/pataylo/cth/cth_6_6_2014/SkyBSrDam2/cth/data/VP_data'

*Bone:
  matep 4  eppvm=user  yield=0.95e9  poisson=0.22  tmelt=1.0

```

¹³Also gives default initial density if material is not porous. If the material is porous, parameter `rp` specifies initial density and `r0` is ambient density for nonporous solid.

jfrac=user	jfpf0=-0.775e9	jfd1=0.100	jfd2=0.0
jfd3=0	jfd4=0	jfd5=0	jftm=1.e20

5.2. **White Matter.**

5.3. **Gray Matter.**

5.4. **Cerebral Spinal Fluid (CSF).**

5.5. **Dry Air.**

6. ALEGRA IMPLEMENTATION

Need to match material models in CTH to ALEGRA.

7. NOTES

- From Taylor 2009, “A typical blast simulation required 31 h using 64 processors on the Sandia National Laboratories Thunderbird computer to integrate out to a time of 2 ms for a 16.72×10^6 cell calculation.”

8. QUESTIONS FOR PAUL

- Why keep the critical value of equivalent plastic strain inside the time interval, or at least confirm this “critical value” is time-independent.
- Do we ever see fully damaged material $D = 1$ in our sims?
- If we have fully damaged material, do we have a loss of ellipticity for the strength equations?
- What strain tensor is being described the equivalent plastic strain?
- Is Mie-Grüneisen equation of state restricted to the use of solids only, or can it be applied to liquids and gases too?

APPENDIX A. DETAILS ON MATERIAL MODELS

A.1. Mie-Grüneisen Equation of State. The Mie-Grüneisen equation of state (MGEOS) is relationship between pressure P and volume V of a solid [ask Paul if just a solid, or applicable to liquid and gas too?] at a given temperature T . The work was originally published by Mie¹⁴ in 1903 and then later in Grüneisen¹⁵ in 1912.

The development of the MGEOS starts from the Grüneisen model that states the Grüneisen parameter Γ , a parameter that describes thermal pressure from a set of vibrating atoms, is related to pressure P and internal energy U through

$$\Gamma = V \left(\frac{dP}{dU} \right)_V. \quad (6)$$

Assume, next, that the Grüneisen parameter Γ is independent of pressure P and internal energy U . Then the foregoing equation can be rewritten as

$$dP = \frac{\Gamma}{V} dU, \quad (7)$$

$$\int dP = \frac{\Gamma}{V} \int dU. \quad (8)$$

$$(9)$$

Next, integrate the foregoing equation from P_0 and U_0 , which exist at the 0 reference state, usually assumed to be absolute freezing, 0° Kelvin, Assume P_0 and U_0 are independent of of temperature T and that pressure P and internal energy U can be estimated from the Hugoniot equations. This gives

$$P - P_0 = \frac{\Gamma}{V} (U - U_0). \quad (10)$$

Next, consider the Hugoniot equations for conservation of mass, momentum, and energy. The Hugoniot equation for conservation of mass relates reference density ρ_0 , density due to shock compression ρ , shock velocity v_s , and particle velocity v_p , as

$$\rho_0 v_s = \rho (v_s - v_p). \quad (11)$$

¹⁴Mie G. Zur kinetischen Theorie der einatomigen Krper. Annalen der Physik. 1903 Jan 1;316(8):657-97.

¹⁵Grüneisen E. Theorie des festen Zustandes einatomiger Elemente. Annalen der Physik. 1912 Jan 1;344(12):257-306.

APPENDIX B. VECTOR ALGEBRA AND CALCULUS NOTES

As defined by Gurtin,¹⁶

- LIN= the set of all tensors
- LIN⁺= the set of all sensors $\mathbf{S}$ with $\det \mathbf{S} > 0$
- SYM= set of all symmetric tensors
- SKW= set of all skew tensors
- PSYM= set of all positive definite, symmetric tensors
- ORTH= set of all orthogonal tensors
- ORTH⁺= set of all rotations

As defined by Curnier,¹⁷ for all second-order tensors $\mathbf{S} \in \text{LIN}$, the **tracor** $\mathbb{T}$ and **deviator** $\mathbb{D}$ are fourth-order tensors are defined as

$$\mathbb{T} \triangleq \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad \text{s.t.} \quad \mathbb{T}:\mathbf{S} = \mathbf{S}_{\text{vol}}, \quad (12)$$

and

$$\mathbb{D} \triangleq \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad \text{s.t.} \quad \mathbb{D}:\mathbf{S} = \mathbf{S}_{\text{dev}}, \quad (13)$$

¹⁶Gurtin ME. An introduction to continuum mechanics. Academic press; 1982, Vol. 158, Jan 12.

¹⁷Curnier A. Computational methods in solid mechanics. Springer Science & Business Media; 2012 Dec 6.